

DVP-SX2

Instruction Sheet

安 裝 說 明
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Multi-Functional, Built-in Analog I/O, Multiple Instructions

多功能內建類比 I/O 豐富指令集

多功能內建模擬量 I/O 豐富指令集

DVP-0150030-02
20240221

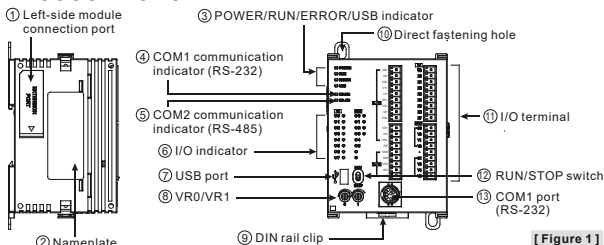


■ ENGLISH ■

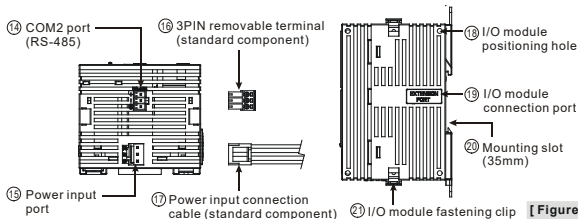
Thank you for choosing Delta DVP-SX2. DVP-SX2 is a 20-point (8DI + 6 DO + 4AI + 2AO) PLC MPU, offering various instructions and is with 16k steps program memory, able to connect with all Slim series extension models, including digital input/output (max. 480 input/output extension points), analog modules (A/D, D/A transformation and temperature units) and all kinds of new high-speed extension modules. Its 2-group high-speed (100kHz) pulse outputs and the one new 2-axis interpolation instructions satisfy all kinds of applications. DVP-SX2 is small in size, and it can be installed easily. Users do not have to install any batteries in DVP-SX2 series PLCs. The PLC programs and the latched data are stored in the high-speed flash memories.

- EN ✖ DVP-SX2 is an OPEN-TYPE device. It should be installed in a control cabinet free of airborne dust, humidity, electric shock and vibration. To prevent non-maintenance staff from operating DVP-SX2, or to prevent an accident from damaging DVP-SX2, the control cabinet in which DVP-SX2 is installed should be equipped with a safeguard. For example, the control cabinet in which DVP-SX2 is installed can be unlocked with a special tool or key.
- EN ✖ DO NOT connect AC power to any of I/O terminals, otherwise serious damage may occur. Please check all wiring again before DVP-SX2 is powered up. After DVP-SX2 is disconnected, Do NOT touch any terminals in a minute. Make sure that the ground terminal (⚡) on DVP-SX2 is correctly grounded in order to prevent electromagnetic interference.
- FR ✖ DVP-SX2 est un module OUVERT. Il doit être installé que dans une enceinte protectrice (boîtier, armoire, etc.) saine, dépourvue de poussière, d'humidité, de vibrations et hors d'atteinte des chocs électriques. La protection doit éviter que les personnes non habilitées à la maintenance puissent accéder à l'appareil (par exemple, une clé ou un outil doivent être nécessaire pour ouvrir la protection).
- FR ✖ Ne pas appliquer la tension secteur sur les bornes d'entrées/Sorties, ou l'appareil DVP-SX2 pourra être endommagé. Merci de vérifier encore une fois le câblage avant la mise sous tension du DVP-SX2. Lors de la déconnection de l'appareil, ne pas toucher les connecteurs dans la minute suivante. Vérifier que la terre est bien reliée au connecteur de terre (⚡) afin d'éviter toute interférence électromagnétique.

■ Product Profile



[Figure 1]



[Figure 2]

■ Electrical Specifications

Model	DVP20SX211R	DVP20SX211T	DVP20SX211S
Item			
Power supply voltage	24VDC (-15% ~ 20%) (with counter-connection protection on the polarity of DC input power) DVPPS01(PS02): input 100-240VAC, output 24VDC/1A(PS02: 2A)		
Inrush current	Max. 7.5A@24VDC		
Fuse capacity	2.5A/30VDC, Polyswitch		
Power consumption	4.7W	4W	4W
Insulation resistance	> 5MΩ (all I/O point-to-ground: 500VDC)		
Noise immunity	ESD (IEC 61131-2, IEC 61000-4-2): 8kV Air Discharge EFT (IEC 61131-2, IEC 61000-4-4): Power Line: 2kV, Digital I/O: 1kV, Analog & Communication I/O: 1kV RS (IEC 61131-2, IEC 61000-4-3): 26MHz ~ 1GHz, 10V/m Surge(IEC 61131-2, IEC 61000-4-5): DC power cable: differential mode ± 0.5 kV analog I/O, RS-232, USB (shielded): Common mode ± 1 kV digital I/O, RS-485 (unshielded): Common mode ± 1 kV		
Grounding	The diameter of grounding wire cannot be smaller than the wire diameter of terminals 24V and 0V (All DVP units should be grounded directly to the ground pole).		
Operation / storage	Operation: 0°C ~ 55°C (temp.), 50 ~ 95% (humidity), Pollution degree2 Storage: -25°C ~ 70°C (temp.), 5 ~ 95% (humidity)		
Vibration / shock resistance	International standards: IEC61131-2, IEC 68-2-6 (TEST Fc)/IEC61131-2 & IEC 68-2-27 (TEST Ea)		
Weight (g)	243g	224g	227g

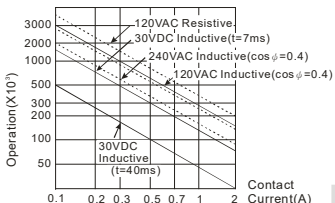
Spec.	Input Point		
Items	24VDC (-15% ~ 20%) single common port input		
Input No.	X0, X2	X1, X3	X4 ~ X7
Input type	DC (SINK or SOURCE)		
Input Current ($\pm 10\%$)	24VDC, 5mA		
Input impedance	4.7K Ohm		
Action level	Off→On	> 15VDC	
	On→Off	< 5VDC	
Response time	Off→On	< 2.5μs	< 10μs
	On→Off	< 5μs	< 20μs
Filter time	Adjustable within 0 ~ 20ms by D1020 (Default: 10ms)		

Spec.	Output Point			
Items	Relay	Transistor		
Output No.	Y0 ~ Y5	Y0, Y2	Y1, Y3	Y4, Y5
Max. frequency	1Hz	100kHz	10kHz	1kHz
Working voltage	250VAC, < 30VDC	5 ~ 30VDC #1		
Max. load	Resistive	SX211T: 0.5A/1 point (3A/ZP) SX211S: 0.3A/1 point (1.8A/UP)		
	Inductive	#2		
	Lamp	15W (30VDC) 2.5W (30VDC)		
Response time	Off→On	Approx. 10 ms	2μs	20μs
	On→Off	Approx. 10 ms	3μs	30μs

#1: DVP20SX211T: UP, ZP must work with external auxiliary power supply 24VDC (-15% ~ +20%), rated consumption approx. 3mA/point.

DVP20SX211S: UP, ZP must work with external auxiliary power supply 5~30VDC, rated consumption approx. 5mA/point.

#2: Life curves



[Figure 3]

A/D and D/A Specifications

Items	Analog Input (A/D)			Analog Output (D/A)		
	Voltage	Current		Voltage	Current	
Analog I/O range	±10V	±20mA	4 ~ 20mA ^{#1}	±10V	0 ~ 20mA	4 ~ 20mA ^{#1}
Digital conversion range	±2,000	±2,000	0 ~ +2,000	±2,000	0 ~ +4,000	0 ~ +4,000
Resolution ^{#2}	12-bit					
Input impedance	> 1MΩ	250Ω		-		
Tolerance carried impedance	-			≥ 5KΩ	≤ 500Ω	
Overall accuracy	Non-linear accuracy: ±1% of full scale within the range of PLC operation temperature Maximum deviation: ±1% of full scale at 20mA and +10V					
Response time	2ms (set up in D1118) ^{#3}			2ms ^{#4}		
Absolute input range	±15V	±32mA		-		
Digital data format	2's complement of 16-bit, 12 significant bits					
Average function	Provided (set up in D1062) ^{#5}			-		
Isolation method	No Isolation between digital circuit and analog circuit					
Protection	Voltage output has short circuit protection, but a long period of short circuit may cause internal wire damage and open circuit of current output.					

#1: Please refer to the detailed explanation of D1115.

#2: Resolution formula

Analog Input (A/D)		Analog Output (D/A)	
Voltage	Current	Voltage	Current
$(5\text{mV} = \frac{20\text{V}}{4000})$	$(10\mu\text{A} = \frac{40\text{mA}}{4000})$	$(5\text{mV} = \frac{20\text{V}}{4000})$	$(5\mu\text{A} = \frac{20\text{mA}}{4000})$

#3: When the scan period is longer than 2ms or the set value, the setting will follow the scan period.

#4: When the scan period is longer than 2ms, the setting will follow the scan period.

#5: When the sampling range is "1", the present value will be read.

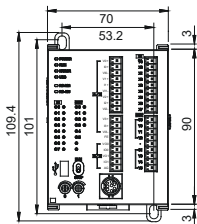
I/O Configuration

Model	Input		Output		I/O configuration					
	Point	Type	Point	Type	Relay		NPN		PNP	
20SX211R	8	DC (Sink Or Source)	6	Relay	V0+	S/S	V0+	S/S	V0+	S/S
					I0+	X0	I0+	X0	I0+	X0
					V10-	X1	V10-	X1	V10-	X1
					V1+	X2	V1+	X2	V1+	X2
					I1+	X3	I1+	X3	I1+	X3
					V11-	X4	V11-	X4	V11-	X4
20SX211T	8	DC (Sink Or Source)	6	NPN Transistor	V2+	X5	V2+	X5	V2+	X5
					I2+	X6	I2+	X6	I2+	X6
					V12-	X7	V12-	X7	V12-	X7
					V3+	C0	V3+	UP	V3+	UP
					I3+	Y0	I3+	ZP	I3+	ZP
					V13-	Y1	V13-	Y0	V13-	Y0
20SX211S	8	DC (Sink Or Source)	6	PNP Transistor	FE	Y2	FE	Y1	FE	Y1
					VO0	●	VO0	Y2	VO0	Y2
					IO0	C1	IO0	Y3	IO0	Y3
					VO1	Y3	VO1	Y4	VO1	Y4
					IO1	Y4	IO1	Y5	IO1	Y5
					AG	Y5	AG	●	AG	●
SX2-R/T/S	4	Analog Input	2	Analog output						

Dimension & Installation

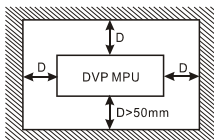
Please install the PLC in an enclosure with sufficient space around it to allow heat dissipation, as shown in the [Figure 5].

- **Direct Mounting:** Please use M4 screw according to the dimension of the product.
- **DIN Rail Mounting:** When mounting the PLC to 35mm DIN rail, be sure to use the retaining clip to stop any side-to-side movement of the PLC and reduce the chance of wires being loose. The retaining clip is at the bottom of the PLC. To secure the PLC to DIN rail, pull down the clip, place it onto the rail and gently push it up. To remove the PLC, pull the retaining clip down with a flat screwdriver and gently remove the PLC from DIN rail.



Unit: mm

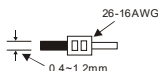
[Figure 4]



[Figure 5]

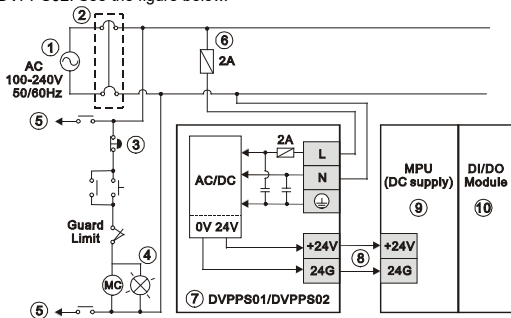
■ Wiring

1. Use 26-16AWG (0.4~1.2mm) single or multiple core wire on I/O wiring terminals. See the figure in the right hand side for its specification. PLC terminal screws should be tightened to 2.0 kg-cm (1.77 in-lbs) and please use only 60/75°C copper conductor.
2. DO NOT wire empty terminal and place the I/O signal cable in the same wiring circuit.
3. DO NOT drop tiny metallic conductor into the PLC while screwing and wiring. Tear off the sticker on the heat dissipation hole for preventing alien substances from dropping in, to ensure normal heat dissipation of the PLC.



◆ Safety Wiring

Since DVP-SX2 is only compatible with DC power supply, Delta power supply modules (DVPPS01/DVPPS02) are suitable power supplies for DVP-SX2. Users are suggested to install the protection circuit at the power supply terminal to protect DVPPS01 or DVPPS02. See the figure below.



[Figure 6]

- | | |
|---|----------------------------------|
| ① AC power supply: 100 ~ 240VAC, 50/60Hz | ② Breaker |
| ③ Emergency stop: This button cuts off the system power supply when accidental emergency takes place. | |
| ④ Power indicator | ⑤ AC power supply load |
| ⑥ Power supply circuit protection fuse (2A) | ⑦ DVPPS01/DVPPS02 |
| ⑧ DC power supply output: 24VDC, 500mA | ⑨ DVP-PLC (main processing unit) |
| ⑩ Digital I/O module | |

◆ Power Supply

The power input of DVP-SX2 series is DC. When operating DVP-SX2 series, please note the following points:

1. The power is connected to the two terminals, 24VDC and 0V, and the range of power is 20.4 ~ 28.8VDC. If the power voltage is less than 20.4VDC, PLC will stop running, all outputs will go "Off" and ERROR indicator will flash continuously.
2. The power shutdown of less than 10 ms will not affect the operation of the PLC. However, power shutdown time that is too long or the drop of power voltage will stop the operation of the PLC and all outputs will go OFF. When the power returns to normal status, the PLC will automatically resume operation. (Care should be taken on the latched auxiliary relays and registers inside the PLC when programming).

◆ Input Point Wiring

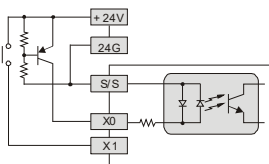
There are 2 types of DC inputs, SINK and SOURCE. (See the example below. For detailed point configuration, please refer to the specification of each model.)

• DC Signal IN – SINK mode

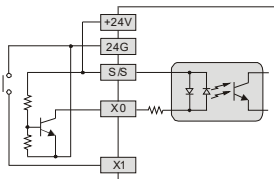
Input point loop equivalent circuit

• DC Signal IN – SOURCE mode

Input point loop equivalent circuit



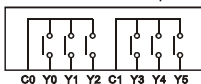
[Figure 7]



[Figure 8]

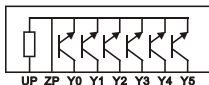
◆ Output Point Wiring

1. DVP-SX2 series have three output modules, relay and transistor (NPN/PNP). Be aware of the connection of shared terminals when wiring output terminals.
2. Output terminals, Y0, Y1, and Y2, of relay models use C0 common port; Y3, Y4, and Y5 use C1 common port; as shown in the [Figure 9]. When output points are enabled, their corresponding indicators on the front panel will be on.

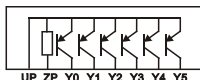


[Figure 9]

3. The output terminals Y0~Y5 of the transistor (NPN) model are connected to the common terminals UP and ZP. See [Figure 10a]. The output terminals Y0~Y5 on the transistor (PNP) model are connected to the common terminals UP and ZP. See [Figure 10b].



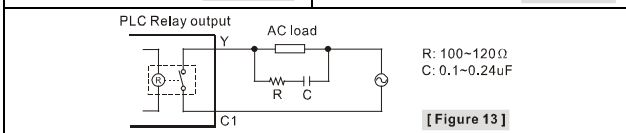
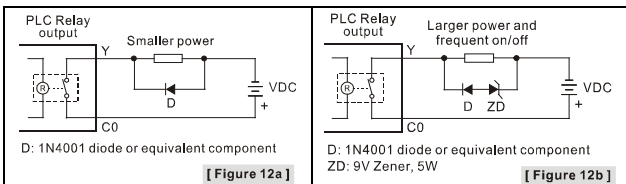
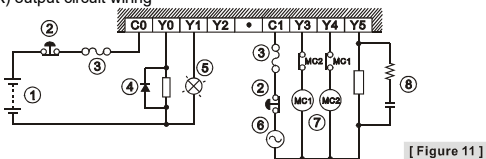
[Figure 10a]



[Figure 10b]

4. Isolation circuit: The optical coupler is used to isolate signals between the circuit inside PLC and input modules.

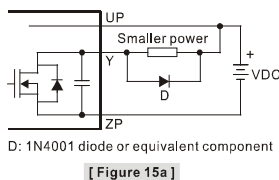
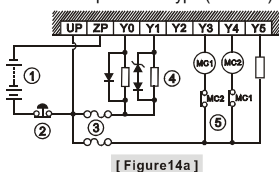
• Relay (R) output circuit wiring



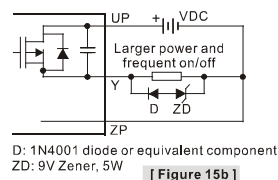
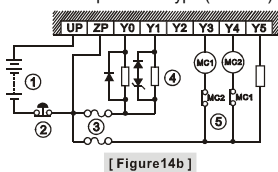
- ① DC power supply
- ② Emergency stop: Uses external switch
- ③ Fuse: Uses 5~10A fuse at the shared terminal of output contacts to protect the output circuit
- ④ Transient voltage suppressor (SB360 3A 60V): To extend the life span of contact.
 1. Diode suppression of DC load: Used in smaller power (Figure 12a)
 2. Diode + Zener suppression of DC load: Used when in larger power and frequent On/Off (Figure 12b)
- ⑤ Incandescent light (resistive load)
- ⑥ AC power supply
- ⑦ Manually exclusive output: For example, Y3 and Y4 control the forward running and reverse running of the motor, forming an interlock for the external circuit, together with the PLC internal program, to ensure safe protection in case of any unexpected errors.
- ⑧ Absorber: To reduce the interference on AC load (Figure 13)

• Transistor (T) output circuit wiring

Transistor output - NPN type (SX211T):



Transistor output - PNP type (SX211S):



- ① DC power supply
- ② Emergency stop
- ③ Circuit protection fuse

